QISKIT FALL FEST 2026

TITLE PAGE



- **Problem Statement ID – VNQFF-08**
- **Problem Statement Title – Quantum-Enhanced Last-Mile Route Optimisation**
- **Team Name – Quantum Sparks (Team ID: HT55)**
- **Our Idea – SparkRoute, a hybrid classical–quantum route optimiser**

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SPARKROUTE

❖ Proposed Solution: Hybrid Classical + QAOA Routing

Detailed explanation

- One depot (A) and 3 delivery stops (B, C, D), linked by a 4 × 4 Euclidean distance matrix
- The visiting order becomes a QUBO with 9 binary variables and is solved with QAOA in Qiskit

How it addresses the problem

- Choices explode with size: 3 stops give 6 orders, 4 give 24, 12 give 479,001,600
- Rules (one stop per position) are folded into the cost as penalties (penalty 20)

Innovation and uniqueness

- Classical exhaustive search stays as the reference for every quantum run
- The QAOA answer is decoded, validated and compared before it is accepted

SparkRoute prototype at a glance

4 locations: depot + 3 delivery stops	**9** qubits (3 stops × 3 positions)
16.01 best classical route distance	**0%** QAOA gap to classical best (16.01 vs 16.01)

Local statevector simulator. No quantum advantage claimed.

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TECHNICAL APPROACH

Technologies

Python 3 · Qiskit · qiskit-optimization · qiskit-algorithms · NumPy · Pandas · Matplotlib

Methodology and process

CLASSICAL

1 Input
Depot A + stops B, C, D; 4×4 Euclidean distance matrix

2 Baseline
All 3! = 6 orders checked; best distance 16.01

QUANTUM (SIMULATED)

3 QUBO
9 binary variables x[i,t], 30 quadratic terms, penalty 20

4 QAOA
reps = 1, COBYLA 15 iterations, 512 shots

CHECK

5 Decode
Reshape to a 3×3 matrix; one stop per position

6 Compare
QAOA route distance vs classical best

Qubit mapping (9 qubits)

Stop	Position 1	Position 2	Position 3
B	x00 · q0	x01 · q1	**x02 · q2**
C	x10 · q3	**x11 · q4**	x12 · q5
D	**x20 · q6**	x21 · q7	x22 · q8

x[i,t] = 1 means stop i is visited at position t. Highlighted cells are the QAOA result: D, C, B.

Qiskit usage

QuadraticProgram builds the 9-variable routing model

QAOA + COBYLA variational loop, reps = 1, 15 iterations

StatevectorSampler local simulator, 512 shots

MinimumEigenOptimizer solves and returns the 9-bit answer

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FEASIBILITY AND VIABILITY

Feasibility analysis

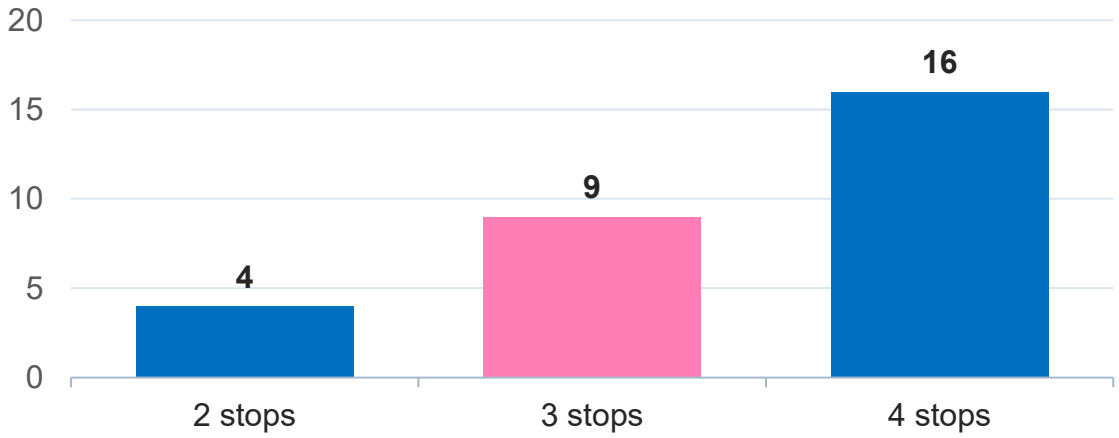
512 bit-strings in the 9-qubit search space (2^9)

6 valid orders, one stop per position (3!)

16.01 best cost, reached by 2 mirror routes

16.01 QAOA route D-C-B

Qubits needed (n^2 encoding)



Prototype runs at 9 qubits. 12 stops in one model would need 144 qubits, so larger sets need splitting.

Challenges and strategies

Challenge	Strategy
Qubits grow as n^2: 4, 9, 16 for 2, 3, 4 stops	Split large delivery sets into small sub-problems, one QAOA run each
QAOA is probabilistic; the run used no fixed seed, so results can vary	Repeat runs; always decode, validate and compare with the classical best
Simulator only; mirror routes tie, so 2 of 6 orders are optimal	No quantum advantage claimed; next: more stops and IBM hardware

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IMPACT AND BENEFITS

Impact on the target audience

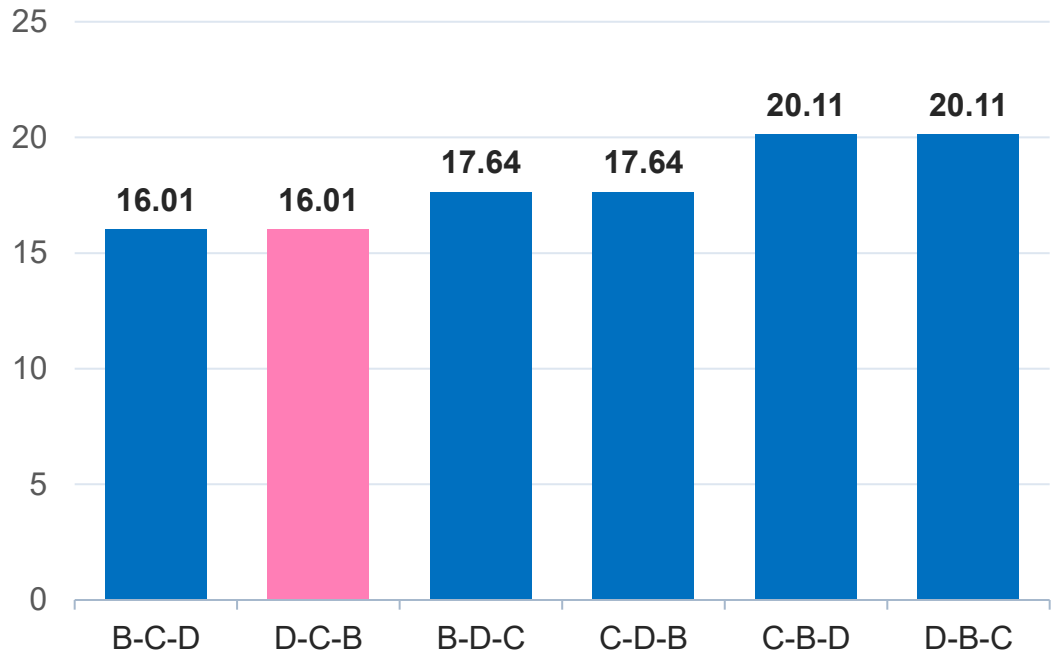
- Fleet planners can test a quantum route-order solver against their classical baseline
- Picking the best order over the worst one cuts this route from 20.11 to 16.01 units (20%)

Benefits

Economic: Shorter routes mean less fuel and driver time

Environmental: Fewer kilometres driven, lower emissions

Technical: The same QUBO pipeline can move to IBM hardware



QAOA chose D-C-B: 16.01, tied for best

Classical best is B-C-D, also 16.01 (difference 0.00).

Simulated coordinates, not road distances; no speedup shown.

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RESEARCH AND REFERENCES

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Research focus VRP · CVRP · VRPTW | Dynamic VRP | QUBO | QAOA | Hybrid quantum–classical

Quantum optimisation

[1] Farhi, Goldstone, Gutmann (2014). A Quantum Approximate Optimization Algorithm. arXiv:1411.4028

[2] Lucas (2014). Ising formulations of many NP problems. Frontiers in Physics 2:5

[3] Harwood et al. (2021). Formulating and Solving Routing Problems on Quantum Computers. IEEE TQE 2:1–17

[4] Osaba et al. (2022). A Systematic Literature Review of Quantum Computing for Routing Problems. IEEE Access

Routing and logistics

[5] Dantzig & Ramser (1959). The Truck Dispatching Problem. Management Science 6(1)

[6] Laporte (1992). The Vehicle Routing Problem: An Overview of Exact and Approximate Algorithms. EJOR 59(3)

[7] Toth & Vigo (2014). Vehicle Routing: Problems, Methods, and Applications. SIAM

[8] INFORMS: UPS ORION project

[9] FarEye last-mile report via DC Velocity

Tools and documentation

[10] Qiskit Optimization docs

[11] Qiskit Algorithms (QAOA)

[12] IBM Quantum Learning

[13] Google OR-Tools: Vehicle Routing

Our claim, kept honest: we test whether hybrid quantum or quantum-inspired search can give better feasible routes for small, structured last-mile problems. We do not claim guaranteed quantum advantage.